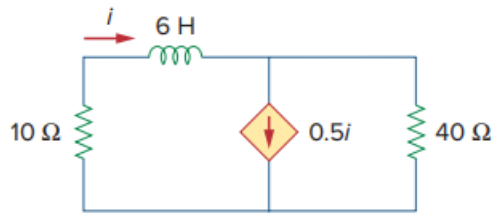


Problem

In the circuit of Fig. 7.99, find $i(t)$ for $t > 0$ if $i(0) = 5$ A.

Fig. 7.99:



Step-by-step solution

Step 1 of 4

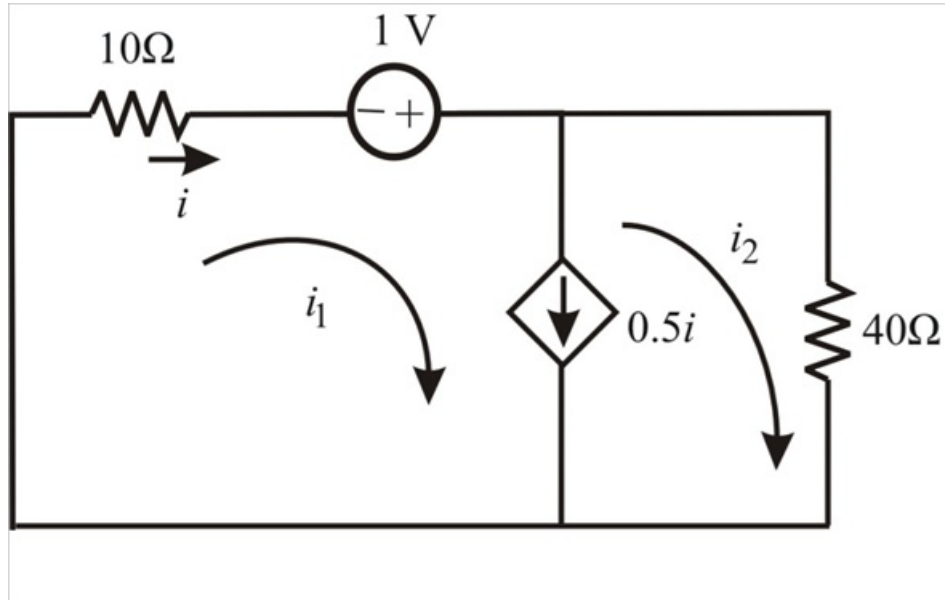
Refer to figure 7.99 in the textbook.

Consider that the circuit is a source free network. Therefore, the current flowing through the inductor is,

$$i(t) = i(0)e^{-\frac{t}{\tau}} u(t)$$

Where $\tau = \frac{L}{R_{eq}}$

Calculate the equivalent resistance R_{eq} . Replace the inductor with 1V voltage source of producing i A current as shown in the following figure:



[Comments \(1\)](#)



Anonymous

Why is the $u(t)$ function in the general equation?

Step 2 of 4

Apply the Kirchhoff's voltage law to the outer loop of the circuit.

$$10i_1 - 1 + 40i_2 = 0 \dots\dots (1)$$

Apply the Kirchhoff's current law.

$$i_1 = i_2 + 0.5i$$

Replace i_1 by i .

$$i = i_2 + 0.5i$$

$$i_2 = 0.5i$$

[Comments \(2\)](#)



Anonymous

Your incorrect, it is $i_1 - i_2 = 0.5i$



Anonymous

Nevermind ...

Step 3 of 4

Recall equation (1).

$$10i_1 - 1 + 40i_2 = 0$$

Replace i_1 by i , and i_2 by $0.5i$.

$$10i - 1 + 40(0.5i) = 0$$

$$i = \frac{1}{30}$$

$$R_{eq} = \frac{1}{i}$$

$$R_{eq} = \frac{1}{\frac{1}{30}}$$

$$R_{eq} = 30 \, \Omega$$

Step 4 of 4

Calculate the time constant.

$$\tau = \frac{6}{30}$$

$$\tau = 0.2 \text{ sec}$$

Therefore, the current flowing through the inductor is,

$$i(t) = 5e^{-\frac{t}{0.2}} u(t) \text{ A}$$

$$i(t) = 5e^{-5t} u(t) \text{ A}$$

Current through inductor for the given circuit is $i(t) = 5e^{-5t} u(t) \text{ A}$.